Viewing Phone on Linux (from Android or iPhone)

Viewing Android/iPhone on Linux

Mirroring an Android phone to a Raspberry Pi using scrcpy

Goal

Mirror the screen of an Android phone onto a Raspberry Pi so you can view and interact with any application on a larger display.

Examples of applications include:

- reading apps
- PDF viewers
- web browsers
- programming apps
- messaging or collaboration tools

Important note

This method mirrors the **phone screen**. It does **not** copy files from the phone automatically.

What appears on the phone screen is transmitted as a video stream to the Raspberry Pi.

What you need

- a Raspberry Pi with Raspberry Pi OS or another Debian-based Linux system
- an Android phone
- a USB cable that supports **data transfer** (not just charging)
- Developer Options enabled on the phone
- USB debugging enabled on the phone

Part 1: Install the required software on the Raspberry Pi

Open a terminal and run:

```
sudo apt update
sudo apt install adb scrcpy
```

Part 2: Enable Developer Options on the phone

If Developer Options are not already enabled:

1. Open `Settings`
2. Go to `About phone`
3. Find `Build number`
4. Tap it 7 times
5. Enter your PIN if prompted

On some phones, `Build number` is inside:

- `Settings -> About phone -> Software information -> Build number`

Part 3: Enable USB debugging on the phone

1. Open `Settings`
2. Go to `Developer options`
3. Turn on `USB debugging`

Depending on the phone, Developer Options may be under:

- `Settings -> System -> Developer options`
- `Settings -> Additional settings -> Developer options`
- `Settings -> System -> Advanced -> Developer options`

Part 4: Connect the phone to the Raspberry Pi

1. Plug the phone into the Raspberry Pi with a USB cable
2. Unlock the phone
3. If prompted, choose:
4. `File Transfer`
5. or `MTP`
6. If the phone asks:

```
Allow USB debugging?
```

tap:

```
Allow
```

Part 5: Check that the Raspberry Pi can see the phone

Run:

```
adb devices -l
```

Expected output should include a line ending in:

```
device
```

Example:

```
List of devices attached
R58N1234567    device usb:1-1 product:... model:... device:...
```

If you see:

```
unauthorized
```

then look at the phone and approve the USB debugging request.

If you see no devices, try:

- reconnecting the cable
- unlocking the phone again
- switching USB mode to `File Transfer`
- using a different cable

Part 6: Mirror the phone screen

Run:

```
scrcpy
```

A window should open on the Raspberry Pi showing the phone screen.

Open any application on the phone and it will appear in the mirrored window.

Part 7: Improve display quality

If the image looks blurry, try increasing bitrate and disabling scrcpy downscaling:

```
scrcpy --max-size 0 --bit-rate 32M
```

You can also rotate the phone to landscape for better viewing.

Part 8: Useful checks

To see the phone's screen resolution:

```
adb shell wm size
```

To restart the adb connection:

```
adb kill-server
adb start-server
adb devices -l
```

Troubleshooting

Problem: `scrcpy` says "no devices/emulators found"

Run:

```
adb devices -l
```

Check:

- USB debugging is enabled
- phone is unlocked
- cable supports data transfer
- you approved the debugging prompt

Problem: phone only charges

Change USB mode to:

```
File Transfer
```

Problem: text or graphics look blurry

Run:

```
scrcpy --max-size 0 --bit-rate 32M
```

Summary

```
sudo apt update
sudo apt install adb scrcpy
adb devices -l
scrcpy
```

Mirroring an iPhone to a Raspberry Pi (AirPlay)

Goal

Mirror the screen of an iPhone onto a Raspberry Pi so you can view and interact with any application on a larger display.

Examples of applications include:

- reading apps
- PDF viewers
- web browsers
- programming apps
- presentation tools

Unlike Android, iPhones do not support `adb` or `scrcpy`. Instead we use Apple's **AirPlay** screen mirroring.

Both devices must be on the **same WiFi network**.

What you need

- Raspberry Pi running Raspberry Pi OS
- iPhone
- both devices connected to the same WiFi network
- AirPlay receiver software installed on the Raspberry Pi

We will use `UxPlay`.

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Part 1: Install required software on the Raspberry Pi

Update the system:

```
sudo apt update
sudo apt upgrade
```

Install build dependencies:

```
sudo apt install cmake g++ pkg-config libavahi-compat-libdnssd-dev \
                 libplist-dev libssl-dev libavcodec-dev libavformat-dev \
                 libavutil-dev libswscale-dev libavahi-client-dev git
```

Clone the project:

```
git clone https://github.com/FDH2/UxPlay.git
cd UxPlay
```

Build the software:

```
mkdir build
cd build
cmake ..
make
```

Install it:

```
sudo make install
```

—

Part 2: Start the AirPlay receiver

Run:

```
uxplay
```

Leave this terminal running.

The Raspberry Pi now appears as an AirPlay device on the network.

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Part 3: Enable screen mirroring on the iPhone

1. Unlock the iPhone
2. Open **Control Center**
    - Face ID phones: swipe down from top-right
    - older phones: swipe up from bottom
3. Tap:

```
Screen Mirroring
```

1. Select the Raspberry Pi.

After connecting, the iPhone screen will appear on the Raspberry Pi.

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Part 4: Open any application

Open any application on the iPhone.

Examples:

- a reading app
- a PDF viewer
- a browser
- a presentation

Everything displayed on the phone will be mirrored.

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Optional: improve display quality

Run uxplay fullscreen:

```
uxplay -f
```

Or disable vertical sync:

```
uxplay -vsync no
```

—

Troubleshooting

Raspberry Pi does not appear in Screen Mirroring

Check:

- both devices are on the same WiFi network
- `uxplay` is running

Restart if needed:

```
uxplay
```

Mirroring is laggy

Possible causes:

- slow WiFi
- crowded network
- low-power Raspberry Pi

Solutions:

- move closer to router
- reduce network traffic

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Summary

Install once:

```
git clone https://github.com/FDH2/UxPlay.git
cd UxPlay
mkdir build
cd build
cmake ..
make
sudo make install
```

Run:

```
uxplay
```

Then on the iPhone:

```
Control Center → Screen Mirroring → select Raspberry Pi
```

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